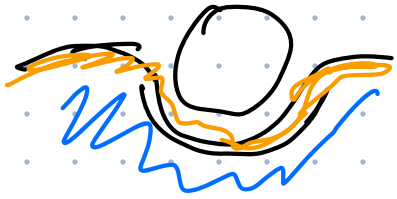
$E =$ ENERGY

VIRIONS] 100 NM

CELL





← BINDING

VOLUME ⇒ PRESSURE



HOW MUCH ENERGY?

- $\sim k_B T$?
- $\sim$ 1 ATP ?

$$E = \iint_{} \sigma \, dA$$ SURFACE

$$= \sigma \cdot A$$ IF CONSTANT

$$E = \iiint \sigma \, dV \qquad \text{VOLUME}$$

IF CONSTANT THEN

$$= \sigma \cdot V$$

$E \overset{?}{=} 1.2 \times 10^{-18}$ J BONDING ?

$= 1256$ pN nm

$= 1.256 \times 10^{3}$ pN nm

$kT \sim 10^{-21}$ J

?? $\Delta E \overset{?}{=} 167000$ pN nm

1.67×10^{5} pN nm